



## Gaussian User Guide

Current as of June 26, 2007

This document is intended to serve as a guide for running Gaussian jobs quickly on ACEnet resources. Please forward any suggestions for its improvement to our tech team at [support@ace-net.ca](mailto:support@ace-net.ca).

If you encounter problems that this document can't resolve, please send a detailed description of your issue to our support team at [support@ace-net.ca](mailto:support@ace-net.ca).

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## 1 Gaussian

ACEnet is pleased to be able to provide access to Gaussian on our Placentia cluster. Gaussian is an electronic structure program that is commonly used by chemists, chemical engineers, biochemists, physicists and others for research in established and emerging areas of chemical interest. More information about Gaussian03 can be found at <http://www.gaussian.com>

Gaussian jobs can be submitted from both `placentia.ace-net.ca` and `placentia2.ace-net.ca`, however interactive jobs can only be run on `placentia2.ace-net.ca`

Users are reminded to keep an eye on interactive jobs and that interactive jobs shouldn't be left to run more than 15 minutes. Jobs longer than this should be run through the N1GE scheduler – see Gaussian Production Jobs.

### 1.1 Setting Gaussian Environment Variables

Once you are logged into the Placentia cluster, you can check your environment variables by typing in 'env' at the command prompt. This will give you a list of all the environment settings that you have enabled for your account.

By default, your environment has **NOT** been set up for Gaussian. You will need to set up the following in your environment and source the following file:

```
(Environment variables – for tcsh)
setenv g03root /usr/local/gaussian
setenv GAUSS_SCRDIR /home/<username>/scratch
```

```
(Gaussian file to source to use the g03 program – for tcsh)
source /usr/local/gaussian/g03/bsd/g03.login
```

```
Now you should see
g03root=/usr/local/gaussian
GAUSS_EXEDIR=/usr/local/gaussian/g03/bsd:/usr/local/gaussian/g03/local:/usr/local/gaussian/g03/extras:/usr/local/gaussian/g03
GAUSS_LEXEDIR=/usr/local/gaussian/g03/linda-exe
GAUSS_ARCHDIR=/usr/local/gaussian/g03/arch
GDVBASIS=/usr/local/gaussian/g03/basis
GAUSS_SCRDIR=/home/<username>/scratch
```

### 1.2 Gaussian Scratch Space

You can see above that the `GAUSS_SCRDIR` is set to `'/home/<username>/scratch'` this means that Gaussian will write all of its temporary files to that directory. Of note – each user has a 200GB quota on their scratch space for placentia – files that are in `/home/<username>/scratch` are under this quota – files that are simply under `/home/<username>/` have a 10GB quota.

Users can customize this by using the command `'setenv GAUSS_SCRDIR /home/<username>/scratch/<directory>'`. This command will now set the Gaussian scratch directory to a directory that is owned and managed by you (<username>) that is within your scratch directory.

This could be useful if you wanted separate directories for the scratch files associated with each Gaussian job you are running.

**NOTE:** be careful when setting your Gaussian scratch directory as some jobs can create very large files (in excess of 80GB). If performance is a consideration and you are using SMALL files, then it might be advantageous to take advantage of a local scratch disk that users can access. The information in this disk is not shared between machines. The total capacity is 63 GB of temporary local space that is available on each compute node.

This is located in `/scratch/tmp` (note the difference `/home/username/scratch` and `/scratch/tmp`)

### 1.3 Automated Environment Configuration

If you would like to have the configurations as shown in 1.1 and 1.2 as your default configuration then you can source our pre-built Gaussian environment set up script. All you need to do is to type in “source /usr/local/lib/gaussian.sh”

### 1.4 Running Gaussian

Gaussian can be run both interactively on **placentia2.ace-net.ca** and through the N1GE batch scheduler. Users are asked to run all jobs through the N1GE batch scheduler, unless they are testing a configuration and wanting to make sure there are no errors prior to submitting a job to the scheduler. Test jobs should not run longer than 15 minutes.

#### 1.4.1 Gaussian Test Jobs

In order to run Gaussian test jobs, users can do this interactively (at the terminal prompt). Before any jobs are run, **remember to set your scratch directory properly and source the proper Gaussian start file (see above)**. Jobs can be executed by running

```
g03 < input > output &
```

Users are again reminded to keep an eye on interactive jobs and that interactive jobs shouldn't be left to run more than 15 minutes. Jobs longer than this should be run through the N1GE scheduler – see Gaussian Production Jobs.

#### 1.4.2 Gaussian Production Jobs

Gaussian production jobs take advantage of the N1GE job scheduler. In order to use the job scheduler users are required to create job submission scripts (see section 7). The following is a sample job submission script for using Gaussian03.

```
#!/bin/sh
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o Gaussian_messages.out
#$ -q linux.q
hostname
date
source /usr/local/lib/gaussian.sh
limit datasize unlimited
limit stacksize 10240 kbytes
limit memoryuse unlimited
limit vmemoryuse unlimited
g03 < ./test001.com > test001_Linux.log
date
```

The above script will run the input file “test001.com” and will output the calculations into the file ‘test001\_Linux.log’. All scratch files will be stored in /home/<username>/scratch and the job will be run with 1 cpu and through the linux.q queue.

All of the output from the job (that is not in the output file) including system messages will be written to Gaussian\_messages.out

## 1.5 Gaussian License Agreement

Users are reminded of the placed on the usage of Gaussian. The ACEnet / Gaussian License Agreement can be downloaded at [http://www.ace-net.ca/software/Gaussian\\_License.pdf](http://www.ace-net.ca/software/Gaussian_License.pdf)

## 2 Test Jobs

Test jobs that require interactive sessions can be directly on the machines. Users are asked to only run short jobs interactively. Users are encouraged to use tools like ps and top to monitor the progress of their test jobs.

## 3. Production Jobs

All production class jobs (ie: not testing) must be submitted via N1 Grid Engine. The commands available in N1GE include:

|        |                                                            |
|--------|------------------------------------------------------------|
| qsub   | submit a batch job to N1 Grid Engine.                      |
| qsh    | submit an interactive X-windows session to N1 Grid Engine. |
| qlogin | submit an interactive login session to N1 Grid Engine.     |
| qrsh   | submit an interactive rsh session to N1 Grid Engine.       |
| qalter | modify a pending batch job of N1 Grid Engine.              |
| qresub | submit a copy of an existing N1 Grid Engine job.           |
| qstat  | show the status of N1 Grid Engine jobs and queues.         |

A detailed user guide of N1GE can be found at <http://docs.sun.com/app/docs/doc/817-6117>.

### 3.1 Gaussian Serial Jobs

For example, one might create the following submit script called `Gaussian_submit.sh` to submit the a serial Gaussian **with no additional memory requirements**:

Here is a sample script called `Gaussian_submit.sh`:

```
#!/bin/sh
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o test_Gaussian_N1GE.out
#$ -l h_vmem=6G
#$ -q linux.q@cl007
hostname
date
source /usr/local/lib/gaussian.sh
limit datasize unlimited
limit stacksize 10240 kbytes
limit memoryuse unlimited
limit vmemoryuse unlimited
g03 < ./test001.com > test001_Linux.log
date
```

To submit the job, type:

```
qsub ./Gaussian_submit.sh
```

The default memory allocated by N1GE is 2GB which equates to 1GB of extra capacity with Gaussian. This is not enough for a number of jobs and thereby users need to know how to modify this setting.

Within the Gaussian input file one might want to change the %MEM setting to use additional memory. The value for %MEM in your Gaussian input file should be 1GB less than the value specified in the N1GE job submission script.

The following submit script called `Gaussian_submit_memory.sh` to submit the a serial Gaussian **with additional memory requirements**:

Here is a sample script called `Gaussian_submit.sh`:

```
#!/bin/sh
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o test_Gaussian_N1GE.out
#$ -l h_vmem=6G
#$ -q linux.q@cl007
hostname
date
source /usr/local/lib/gaussian.sh
limit datasize unlimited
limit stacksize 10240 kbytes
limit memoryuse unlimited
limit vmemoryuse unlimited
g03 < ./test001.com > test001_Linux.out
date
```

**NOTE:** inside test001.com there needs to be a line that says %MEM=5GB (1GB less than the 6GB specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

To submit the job, type:

```
qsub ./Gaussian_submit_memory.sh
```

To view the status of your batch jobs, the `qstat` command can be used:

```
qstat -u <username>
```

where <username> is the username of the user who submitted the job.

### 3.2 Gaussian Parallel Jobs

The submission of parallel jobs to N1GE is similar to the submission of serial jobs, except that the job submission script references the parallel environment in addition to a queue.

The default memory allocated by N1GE is 2GB which equates to 1GB of extra capacity with Gaussian. This is not enough for a number of jobs and thereby users need to know how to modify this setting. Similarly the default number of processors is 1 so users may want to change this also.

It should be noted that your mileage may vary in terms of performance with parallelized Gaussian work, and typically Gaussian does not scale well beyond 8 processors – so think about this when getting your resources.

Within the Gaussian input file one might want to change the %MEM setting to use additional memory. The value for %MEM in your Gaussian input file should be 1GB less than the value specified in the N1GE job submission script.

Within the Gaussian input file there is also the %NPROCS directive that can tell one to use additional processors. This is in addition to the 1 processor already required. So from a N1GE perspective you will need to add one to the %NPROCS value.

For example, one might create the following submit script called `Gaussian_Parallel_submit.sh` to submit a Gaussian parallel job that needs 5 processors and 6GB of Memory per processor:

Here is a sample script called `Gaussian_Parallel_submit.sh`:

```
#!/bin/sh
#$ -S /bin/csh
#$ -N test
#$ -cwd
#$ -j y
#$ -o test_Gaussian_N1GE.out
#$ -l h_vmem=6G
#$ -pe gaussian 5
#$ -q linux.q
hostname
date
source /usr/local/lib/gaussian.sh
limit datasize unlimited
limit stacksize 10240 kbytes
limit memoryuse unlimited
limit vmemoryuse unlimited
g03 < ./test001.com > test001_Linux.out
date
```

**NOTE:** inside test001.com there needs to be a line that says %MEM=5GB (1GB less than the 6GB specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

**NOTE:** inside test001.com there needs to be a line that says %NPROCS=4 (1 less than the 5 processors specified in the job submission script). If this step is not followed precisely then your job will either not run or hang in the job queues.

To submit the job, type:

```
qsub ./Gaussian_Parallel_submit.sh
```

#### 4. Support

ACEnet is in the process of expanding its technical team, however we currently have two High Performance Computing Specialists dedicated to user support (Mark Staveley and Ross Dickson). The best way to access our support team is by sending a detailed description of your problem to [support@ace-net.ca](mailto:support@ace-net.ca). If you need immediate assistance, you can contact:

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